

# Jeongho Kim

✉ jeonghokim@khu.ac.kr    🌐 Website

## Appointments

---

|                                                                                                                      |                              |
|----------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Department of Applied Mathematics, Kyung Hee University</b><br><i>Assistant Professor</i>                         | <i>Sep. 2022 – Present</i>   |
| <b>Center for Mathematical Challenges, Korea Institute for Advanced Study (KIAS)</b><br><i>CMC Fellow</i>            | <i>Mar. 2022 – Aug. 2022</i> |
| <b>Research Institute of Natural Sciences, Hanyang University</b><br><i>Postdoctoral Researcher</i>                  | <i>Mar. 2021 – Mar. 2022</i> |
| <b>Institute for New Media and Communications, Seoul National University (SNU)</b><br><i>Postdoctoral Researcher</i> | <i>Sep. 2019 – Mar. 2021</i> |

## Research Interests

---

- Compressible/incompressible fluid dynamics.
- Kinetic theory of gases.
- Quantum mechanical systems.
- Numerical methods for incompressible fluid/quantum mechanical systems.

## Education

---

|                                                                                                                       |                              |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Seoul National University</b><br><i>Ph.D in Mathematical Sciences</i>                                              | <i>Mar. 2016 – Aug. 2019</i> |
| ◦ Thesis title: : On the fluid coupling and hydrodynamic limit of the kinetic thermomechanical Cucker-Smale equation. |                              |
| ◦ Thesis advisor : <a href="#">Prof. Seung-Yeal Ha.</a> <a href="#">🔗</a>                                             |                              |
| <b>Seoul National University</b><br><i>B.S. in Mathematical Sciences and Economics</i>                                | <i>Mar. 2012 – Feb. 2016</i> |

## Honors, Awards, and Grants

---

|                                                                                        |                              |
|----------------------------------------------------------------------------------------|------------------------------|
| ◦ Research grant supported by National Research Foundation of Korea (Basic Research A) | <i>Sep. 2026 – Aug. 2031</i> |
| ◦ Research grant supported by Samsung Science and Technology Foundation                | <i>Jun. 2024 – May 2027</i>  |
| ◦ Associate member of KIAS                                                             | <i>2023 - Present</i>        |
| ◦ Best doctoral degree thesis award, College of Natural Sciences, SNU                  | <i>2019</i>                  |
| ◦ Grand prize, 9th S-Oil best doctoral degree thesis award                             | <i>2019</i>                  |
| ◦ International Research Training Group (IRTG) 2235                                    | <i>Sep. 2016 – Aug. 2019</i> |
| ◦ Summa Cum Laude, SNU                                                                 | <i>2016</i>                  |
| ◦ Korea National Scholarship for Natural Science and Engineering                       | <i>Mar. 2012 – Feb. 2016</i> |

## Academic Services

---

### Academic Society

- Business Director, Korean Society of Industrial and Applied Mathematics (KSIAM), 2025–2026.
- Ambassador of Mathematical Culture, Korean Mathematical Society (KMS), 2023–2024.

**Journal Referee:** Acta Appl. Math.; Acta Math. Sci.; Adv. Math.; Arch. Rational Mech. Anal.; Appl. Math. Lett.; Chaos; IEEE Trans. Autom. Control; J. Dyn. Differ. Equ.; J. Evol. Eqn.; J. Nonlinear Sci.; J. Stat.

Phys.; Kinet. Relat. Models; Math. Models Methods Appl. Sci.; Netw. Heterog. Media; Physica D; SN PDEA

**Conference organizing**

- (With Kyeongsu Choi and Dohyun Kim) Workshop on Mathematical Physics and Probability: Theory and Application, KIAS, Seoul, Republic of Korea, August 8th – August 10th, 2022.
- Suwon-Incheon workshop, Kyung Hee University, Yongin, Republic of Korea, August 19th, 2024.
- (With Junha Kim) Suwon-Incheon workshop, Courtyard Marriott Suwon, Suwon, Republic of Korea, August 18th–20th, 2025.
- (With Moon-Jin Kang and Bonsuk Kwon) East Asian Workshop on Mathematical Fluid Dynamics and PDEs, Yonsei University, November 3rd–7th, 2026.